Changheng Li

Postdoctoral Researcher · Delft University of Technology

Embedded Systems & WISE · Delft, the Netherlands

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CURRENT POSITION

Postdoctoral Researcher · TU Delft

Research on liquid-crystal reconfigurable intelligent surfaces (LC-RIS), including phase design under slow, asymmetric hardware response and reconfiguration-time constraints.

RESEARCH

Microphone arrays and speech enhancement. Joint estimation of source transfer functions and the power of speech, reverberation and noise, using statistical structure shared across time frames.

Quantized-data inference. Model-order selection from signed measurements; collaborative work on one-bit MIMO channel estimation and radar inference.

Future interests. Communication-and-sensing limits, tracking with slow hardware, and learning methods that retain physical model structure.

Software: MATLAB implementations at TU Delft.

EDUCATION

PhD · Electrical Engineering · 2025

Delft University of Technology

Signal Processing Systems. Supervisors: Alle-Jan van der Veen and Richard C. Hendriks.

MSc · Electronic Engineering · 2017–2020

University of Science and Technology of China

School of Information Science and Technology. Advisor: Jian Li.

BSc · Applied Mathematics · 2013–2017

University of Science and Technology of China

School of the Gifted Young. Advisor: Jian Li.

TEACHING, RECOGNITION AND TOOLS

Teaching: Teaching assistant, Estimation and Detection, TU Delft · 2021/22 Q2.

Award: National Award for Graduates, Ministry of Education, China · 2018.

Tools: MATLAB · Python / PyTorch · LaTeX

Selected publications

Five representative papers. Full publication list · Google Scholar

01 Multimicrophone Signal Parameter Estimation in a Multi-Source Noisy Reverberant Scenario

C. Li and R. C. Hendriks

IEEE Transactions on Audio, Speech and Language Processing, 33, 678–692, 2025.

doi.org/10.1109/TASLPRO.2025.3533374

02 Alternating Least-Squares-Based Microphone Array Parameter Estimation for a Single-Source Reverberant and Noisy Acoustic Scenario

C. Li and R. C. Hendriks

IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31, 3922–3934, 2023.

doi.org/10.1109/TASLP.2023.3306713

03 Joint Maximum Likelihood Estimation of Microphone Array Parameters for a Reverberant Single Source Scenario

C. Li, J. Martinez and R. C. Hendriks

IEEE/ACM Transactions on Audio, Speech, and Language Processing, 31, 695–705, 2023.

doi.org/10.1109/TASLP.2022.3231706

04 Angular-Domain Channel Estimation for One-Bit Massive MIMO Systems: Performance Bounds and Algorithms

F. Liu, H. Zhu, **C. Li**, J. Li, P. Wang and P. V. Orlik

IEEE Transactions on Vehicular Technology, 69, 2928–2942, 2020.

doi.org/10.1109/TVT.2020.2966003

05 Bayesian Information Criterion for Signed Measurements With Application to Sinusoidal Signals

C. Li, R. Zhang, J. Li and P. Stoica

IEEE Signal Processing Letters, 25, 1251–1255, 2018.

doi.org/10.1109/LSP.2018.2847559

PHD THESIS

Multi-Microphone Signal Parameter Estimation in Various Acoustic Scenarios

Delft University of Technology · 2025

Read thesis